Pseudo first order reaction:

Kinetic study of a higher order reaction is difficult to follow, for example, in a study of a second order reaction involving two different reactants; the simultaneous measurement of change in the concentration of both the reactants is very difficult. To overcome such difficulties, A second order reaction can be altered to a first order reaction by taking one of the reactant in large excess, such reaction is called pseudo first order reaction. Let us consider the acid hydrolysis of an ester,

$$CH_3COOCH_3\ (aq) + H_2O\ (l) \xrightarrow{H^+} CH_3COOH\ (aq) + CH_3OH\ (aq)$$

$$\text{Rate} = k\ [CH_3COOCH_3]\ [H_2O]$$

If the reaction is carried out with the large excess of water, there is no significant change in the concentration of water during hydrolysis. i.e.,concentration of water remains almost a constant.

Now, we can define $k\ [H_2O] = k'$; Therefore the above rate equation becomes

$$\text{Rate} = k'\ [CH_3COOCH_3]$$

Thus it follows first order kinetics.

7.5.2 Integrated rate law for a zero order reaction:

A reaction in which the rate is independent of the concentration of the reactant over a wide range of concentrations is called as zero order reactions. Such reactions are rare. Let us consider the following hypothetical zero order reaction.

$A \longrightarrow \text{product}$

The rate law can be written as,

$$\text{Rate} = k\ [A]^0$$

$$\frac{-d[A]}{dt} = k\ (1) \qquad \left(\therefore [A]^0 = 1\right)$$

$$\Rightarrow\ -d[A] = k\ dt$$

Integrate the above equation between the limits of $[A_0]$ at zero time and $[A]$ at some later time 't',

$$-\int_{[A_0]}^{[A]} d[A] = k\int_0^t dt$$

$$-\left([A]\right)_{[A_0]}^{[A]} = k\left(t\right)_0^t$$

$$[A_0] - [A] = kt$$

$$k = \frac{[A_0] - [A]}{t}$$

Equation (2) is in the form of a straight line $y = mx + c$

Ie., $[A] = -kt + [A_0]$

$\Rightarrow y = c + mx$

A plot of $[A]$ Vs time gives a straight line with a slope of $-k$ and y - intercept of $[A_0]$.

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